

Dinesh Behera

PhD Researcher — Perovskite Photovoltaics — Experimental Physicist

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PhD researcher in photovoltaic (PV) materials and device physics, specialising in scalable spray-coated perovskite solar cells. Experienced in cleanroom fabrication, thin-film deposition, device characterisation, and drift-diffusion modelling, with a current focus on integrating perovskite PV onto carbon-fibre-reinforced polymer substrates for lightweight, high-specific-power applications.

TECHNICAL SKILLS

Thin-Film Device Fabrication	N ₂ Glovebox Processing, Spray Coating, Spin Coating, Thermal Evaporation, Electron-Beam Evaporation, Laser Patterning, Cleanroom Processing
Characterisation	Current Density–Voltage (J–V) Measurements, External Quantum Efficiency (EQE), UV–Vis Spectroscopy, Photoluminescence (PL), X-ray Diffraction (XRD), Raman Spectroscopy, Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM), Optical Microscopy, Surface Profilometry, Four-Point Probe
Device Modelling	SCAPS-1D, gpydm, PC1D
Programming & Data Analysis	Python (NumPy, SciPy, Pandas, Matplotlib, pvlib), OriginPro, Microsoft Excel
Scientific Writing & Presentation	LaTeX, Microsoft Word, Google Docs, Microsoft PowerPoint, Google Slides

RESEARCH EXPERIENCE

University of Sheffield	Sheffield, UK
<i>Doctoral Research Project, Supervisor: Prof. David Lidzey</i>	October 2024 – Present

- Fabricate and optimise scalable spray-coated perovskite solar cells on carbon-fibre-reinforced polymer (CFRP) substrates for lightweight, high-specific-power PV applications.
- Fabricate thin-film photovoltaic devices using spray coating, spin coating, N₂ glovebox processing, laser patterning, thermal evaporation, and device encapsulation.
- Characterise photovoltaic devices using J–V measurements, EQE spectroscopy, UV–Vis spectroscopy, PL spectroscopy, XRD, Raman spectroscopy, SEM, AFM, and surface profilometry.
- Investigate the influence of substrate roughness, interfacial engineering and processing conditions, and optimise these parameters to improve thin-film morphology, crystallisation, and device performance.
- Develop fabrication strategies for scaling laboratory-scale perovskite solar cells (0.02 cm²) into series-connected mini-modules (2.38 cm² and 11.62 cm²).

University of Sheffield	Sheffield, UK
<i>MSc Dissertation, Supervisor: Dr. Alex Ramadan</i>	October 2023 – September 2024

- Developed and validated a SCAPS-1D model of a 2.0 eV halide-segregation-suppressed perovskite solar cell based on published experimental work.
- Investigated the influence of absorber, transport layer and interface parameters on open-circuit voltage (V_{OC}), short-circuit current density (J_{SC}), fill factor (FF) and power conversion efficiency (PCE).
- Compared simulated device characteristics with published experimental results to understand the governing physical mechanisms and identify performance-limiting factors.

- Comparative analysis of all-perovskite and III–V multijunction PV for terrestrial and space applications.
- Review of state-of-the-art space PV technologies.
- Investigation of BaZrS₃ chalcogenide perovskites for lead-free tandem and multijunction solar cells.
- Evaluation of A-site mixed-cation lead halide perovskites for improved PV stability and efficiency.

Project Repository: [GitHub](#)

Indian Institute of Technology Indore
Research Training, Supervisor: Prof. Pankaj R. Sagdeo

Indore, India
July 2021 – March 2022

- Investigated the influence of structural disorder on the optical properties of crystalline and amorphous silicon.
- Performed first-principles calculations and spectroscopic analysis to determine the Urbach energy of crystalline and amorphous silicon.
- Correlated Urbach energy with material disorder and evaluated its implications for PV performance.

National Institute of Technology Rourkela
MSc Dissertation, Supervisor: Prof. Pitamber Mahanandia

Rourkela, India
August 2021 – May 2022

- Synthesised SnO₂/reduced graphene oxide (rGO) nanocomposites using solution-based methods.
- Characterised the structural and optical properties using X-ray diffraction (XRD), Raman spectroscopy and UV–Vis spectroscopy.
- Evaluated the potential of the nanocomposites as electron transport layers (ETLs) and photoabsorbers for PV applications.

National Institute of Technology Rourkela
Short-term Research Project, Supervisor: Prof. Pitamber Mahanandia

Rourkela, India
August 2021 – October 2021

- Conducted a comparative study of physical vapour deposition (PVD) techniques, including thermal evaporation, sputtering, ion plating and arc-vapour deposition, for thin-film photovoltaic fabrication.
- Evaluated the suitability of different PVD techniques for depositing functional layers in thin-film solar cells based on material requirements and device architecture.

PROFESSIONAL EXPERIENCE

Ossila Ltd.
Scientific Content Writer

Sheffield, UK
2024

- Authored technical articles on perovskite PV for scientific and industrial audiences. [Author Portfolio](#).
- Communicated complex photovoltaic concepts into clear, accurate, and engaging scientific content.

Chegg Inc.
Advanced Physics Expert

Remote
2022 – 2023

- Solved undergraduate and postgraduate physics problems across multiple disciplines.
- Developed clear, step-by-step technical explanations to enhance students' conceptual understanding and problem-solving skills.

EDUCATION

2024 – Present	Doctor of Philosophy (PhD) in Physics, University of Sheffield	
2023 – 2024	Master of Science in Solar Cell Technology, University of Sheffield	(Distinction)
2017 – 2022	Integrated Master of Science (Physics), National Institute of Technology Rourkela (CGPA: 8.24/10.00; B.Sc.: 8.12, M.Sc.: 8.41)	

KEY ACHIEVEMENTS

- 2026 **Best Poster Award**, Faculty of Science Doctoral School Showcase, University of Sheffield
- 2026 Selected for the **Royal Society of Chemistry (RSC) Broadening Horizons in Chemical Sciences Programme**, Cohort 5
- 2024 Awarded the **EPSRC Doctoral Training Partnership (DTP) Scholarship**, University of Sheffield
- 2023 Awarded the **Postgraduate Scholarship** for the MSc in Solar Cell Technology, University of Sheffield
- 2022 Qualified the **Graduate Aptitude Test in Engineering**, Physics, All India Rank: 2355/19375
- 2017 Qualified the **Joint Entrance Examination (JEE) Main**, All India Rank: 23110/1200000

CONFERENCE & RESEARCH PRESENTATIONS

- 2026 **ISOPHOS 2026 – International School on Hybrid Organic Photovoltaics**, Tuscany, Italy
Oral Presentation: "Toward Structural and Scalable Perovskite PV: Integration on Carbon Fibre Substrates."
- 2026 **Royal Society of Chemistry Chemical Sciences Research Symposium**, Sheffield, UK
Poster Presentation: "Toward Structural Spray-Coated Perovskite PV: Integration on Carbon Fibre Substrates."
- 2026 **UK Semiconductors Conference 2026**, Sheffield, UK
Poster Presentation: "Toward Structural Spray-Coated Perovskite PV: Integration on Carbon Fibre Substrates."
- 2026 **Faculty of Science Doctoral School Showcase 2026**, The University of Sheffield, UK
Poster Presentation: "Integration of Spray-Coated Perovskite Solar Cells onto Carbon Fibre Substrates."

PROFESSIONAL DEVELOPMENT

Royal Society of Chemistry

United Kingdom

Broadening Horizons in the Chemical Sciences – Cohort 5 Scholar

March 2026 – December 2027

- Selected for the Royal Society of Chemistry's year-long Broadening Horizons in the Chemical Sciences professional development programme.
- Participate in mentoring, leadership development, networking, and industry engagement with leading organisations, including RSC, Johnson Matthey, AstraZeneca, GSK, Unilever, Syngenta, Nanomerics, and AM Technology.
- Attend industry site visits, career development workshops, and networking events to strengthen research, communication, and professional skills.

TRAINING & WORKSHOPS

CSIR–Central Electrochemical Research Institute (CECRI)

Karaikudi, India

Training Programme on Energy Materials and Energy Harvesting Devices

June 2021

- Selected for and participated in hands-on training in energy materials, device physics, and fabrication techniques for energy harvesting devices.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

INSPIRE, NIT Rourkela

2019 – 2022

Member

- Organised science outreach workshops, lectures and student engagement activities.

Astro NITR

2017 – 2022

Member

- Organised astronomy outreach events, telescope observation sessions, and public lectures.